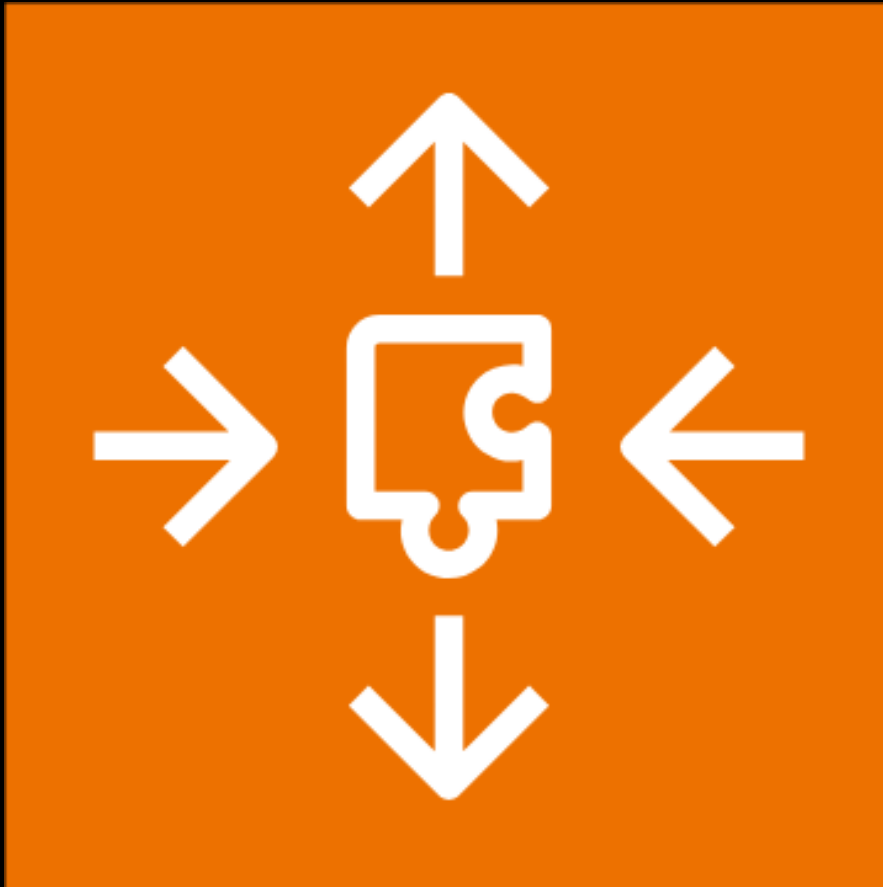
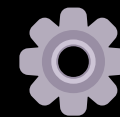




SCALING SMARTLY: AWS AUTO SCALING GROUP (ASG) PROJECT DOCUMENTATION



OVERVIEW

Auto Scaling Groups (ASG) in AWS allow you to automatically scale your EC2 instances based on demand, ensuring high availability and cost efficiency. This guide will walk you through setting up an ASG using a launch template and configuring scaling policies.

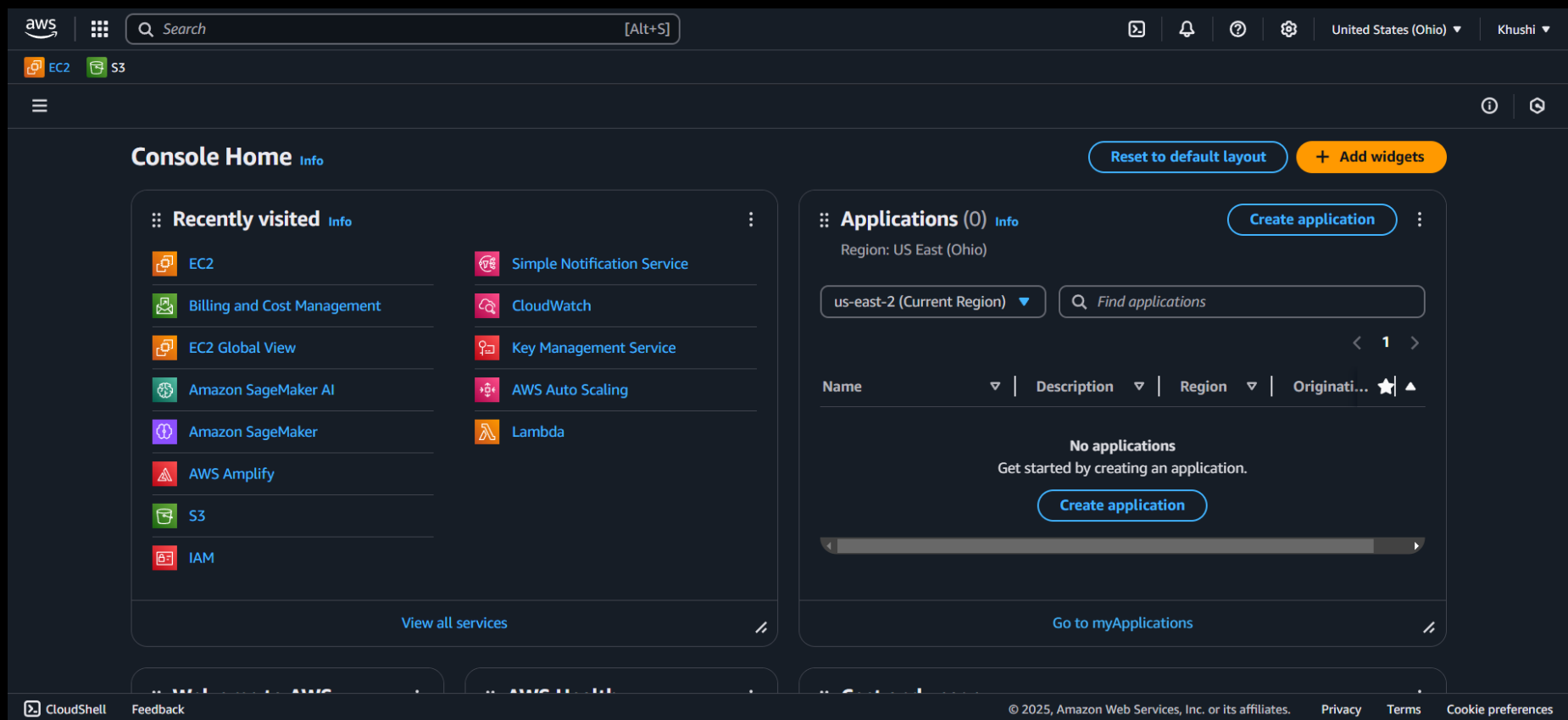
PREREQUISITES

Before proceeding, ensure you have the following:

- ✓ An active AWS account.
- ✓ Basic knowledge of AWS EC2 and Auto Scaling Groups.
- ✓ An IAM user with necessary permissions to create EC2 instances and ASGs.
- ✓ A configured key pair for SSH access (if needed).

I. ACCESSING AWS MANAGEMENT CONSOLE

1. Go to the AWS Management Console.



2. Search for EC2 and navigate to the EC2 dashboard.

The screenshot displays the AWS Management Console interface for the EC2 dashboard. The top navigation bar includes the AWS logo, a search bar, and user information (United States (Ohio) and Khushi). The left sidebar shows navigation links for Dashboard, EC2 Global View, Events, Instances, Images, and Elastic Block Store. The main content area is divided into several sections:

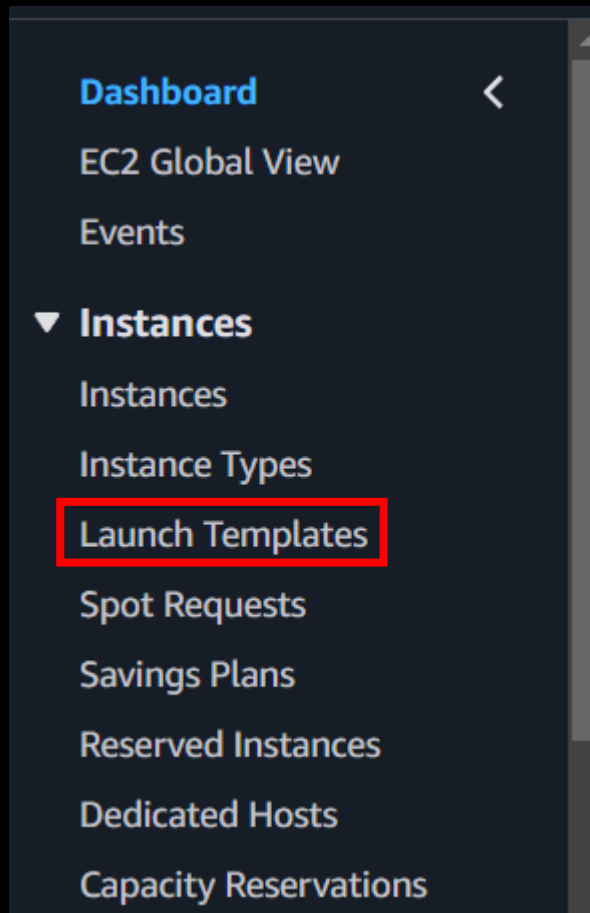
- Resources:** A table showing the following Amazon EC2 resources in the US East (Ohio) Region:

Resource	Count
Instances (running)	0
Auto Scaling Groups	0
Capacity Reservations	0
Dedicated Hosts	0
Elastic IPs	0
Instances	0
Key pairs	3
Load balancers	0
Placement groups	0
Security groups	4
Snapshots	0
Volumes	0
- Launch instance:** A section with a description and buttons to 'Launch instance' and 'Migrate a server'.
- Service health:** A section showing the 'AWS Health Dashboard' and the status of the service (operating normally).
- EC2 Free Tier:** A section indicating that 0 EC2 free tier offers are in use, with a warning that 0 offers are forecasted to exceed the free tier limit.
- Account attributes:** A section showing the 'Default VPC' and 'Settings'.

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II. CREATING A LAUNCH TEMPLATE

1. On the left-hand panel, click on "Launch Templates".



2. Click on "Create launch template".

Compute

EC2 launch templates

Streamline, simplify and standardize instance launches

Use launch templates to automate instance launches, simplify permission policies, and enforce best practices across your organization. Save launch parameters in a template that can be used for on-demand launches and with managed services, including EC2 Auto Scaling and EC2 Fleet. Easily update your launch parameters by creating a new launch template version.

New launch template

Create launch template

Benefits and features

- Streamline provisioning
- Simplify permissions

Documentation

[Documentation](#)

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3. Provide a name for the template (e.g., MyTemp).

 [EC2](#) > [Launch templates](#) > Create launch template

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

MyTemp

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

A prod webserver for MyApp

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► **Template tags**

► **Source template**

4. Scroll down to the AMI (Amazon Machine Image) section in Launch template contents.

Launch template contents
Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ **Application and OS Images (Amazon Machine Image)** Info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

Recents

Quick Start

☒ Don't include in launch template

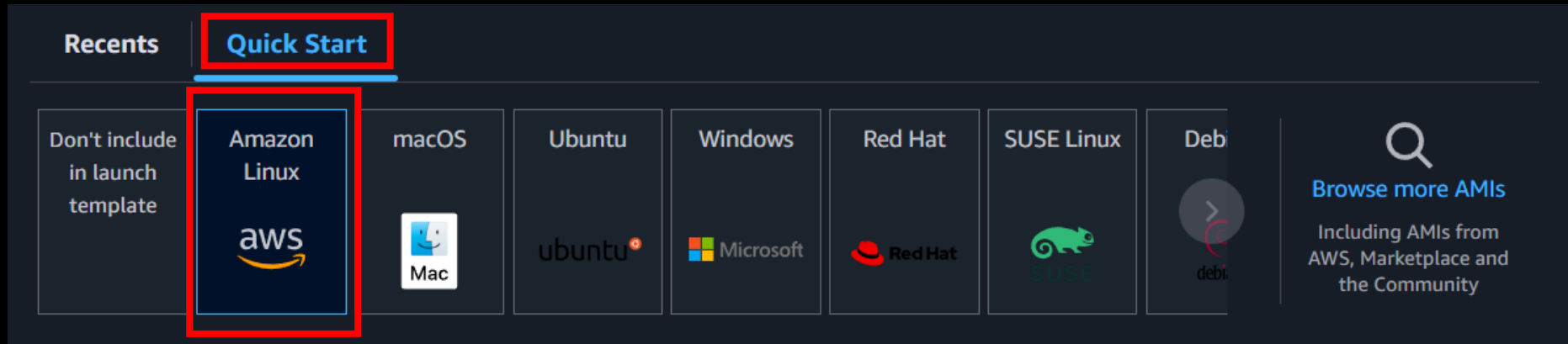
☐ Recently launched

Q

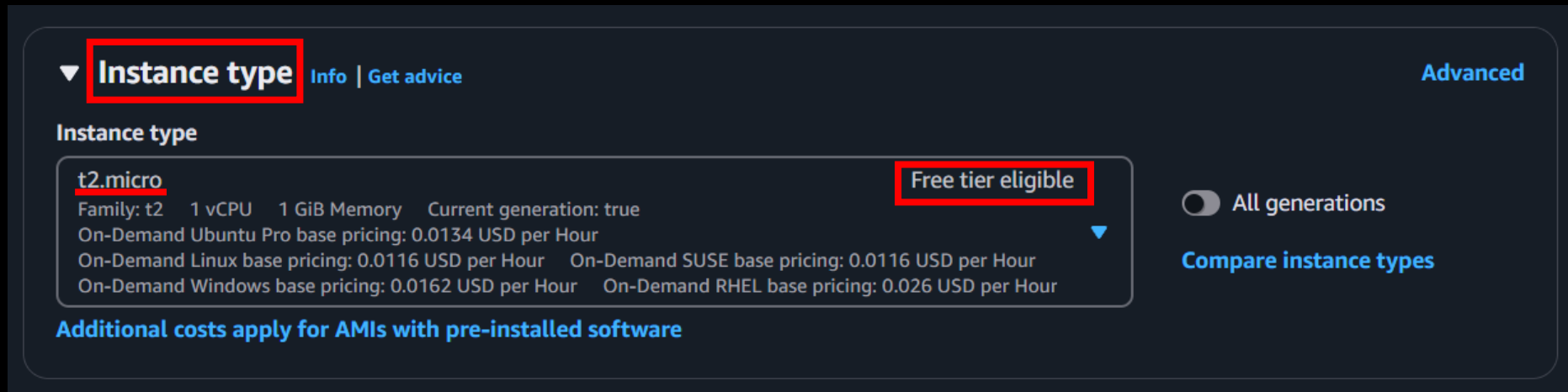
[Browse more AMIs](#)

Including AMIs from AWS, Marketplace and the Community

5. Click on "Quick Start" and choose Amazon Linux.



6. Select the instance type as t2.micro (free tier eligible).



7. Include a Key Pair for SSH access.

▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

key-win

▼

 [Create new key pair](#)

8. Keep the rest of the setting and same and In Summary click on "Create launch template".

▼ Summary

Software Image (AMI)
Amazon Linux 2023 AMI 2023.6.2...[read more](#)
ami-018875e7376831abe

Virtual server type (instance type)
t2.micro

Firewall (security group)
-

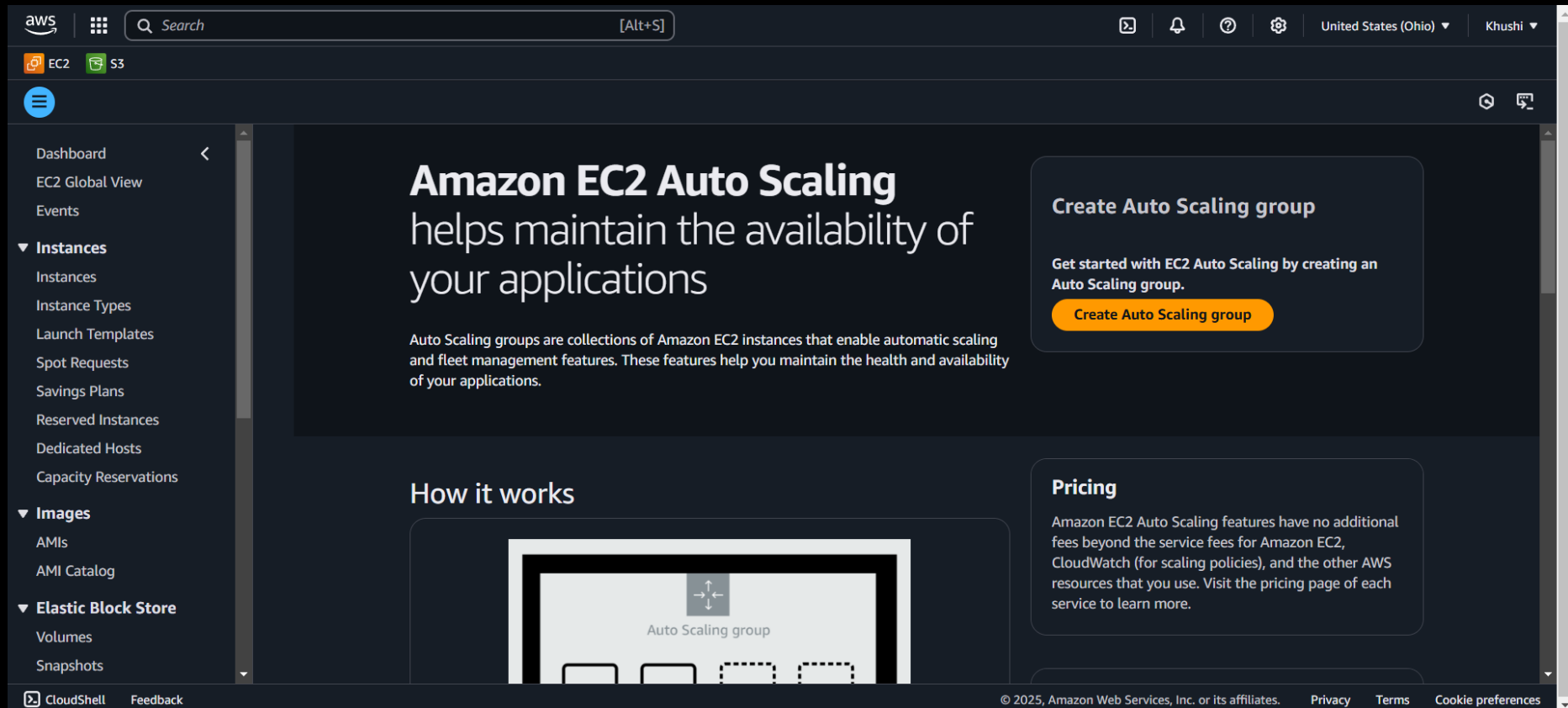
Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year includes 750 hours of t2.micro (or t3.micro in the Regions in which t2.micro is unavailable) instance usage on free tier AMIs per month, 750 hours of public IPv4 address usage per month, 30 GiB of EBS storage, 100 GB of S3 storage, and 100 GB of ElastiCache storage.

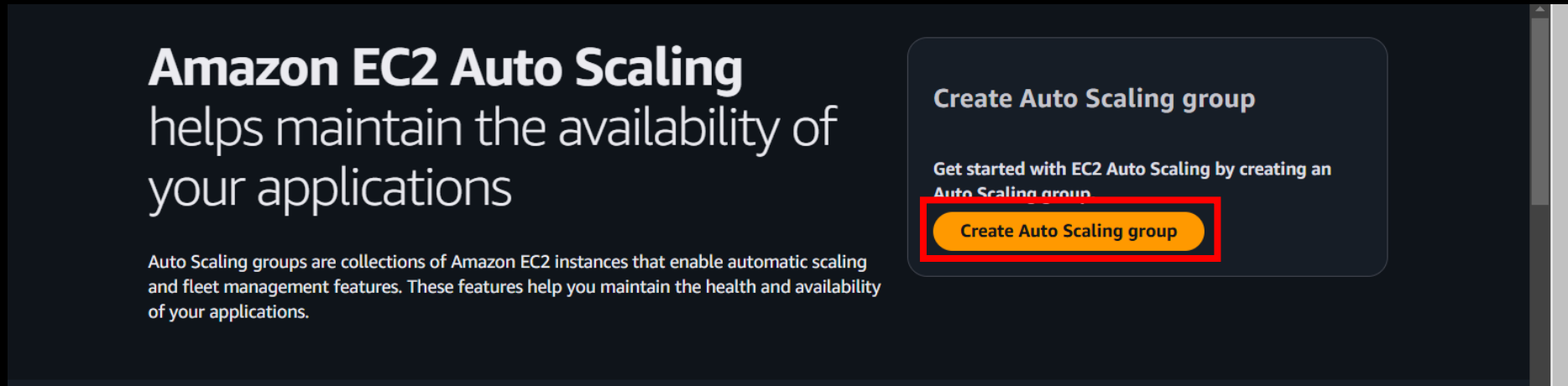
[Cancel](#) [Create launch template](#)

III. SETTING UP AUTO SCALING GROUP (ASG) 🏛️

1. Navigate to ASG (Auto Scaling Groups) in the AWS console.



2. Click on "Create Auto Scaling Group".



Amazon EC2 Auto Scaling
helps maintain the availability of
your applications

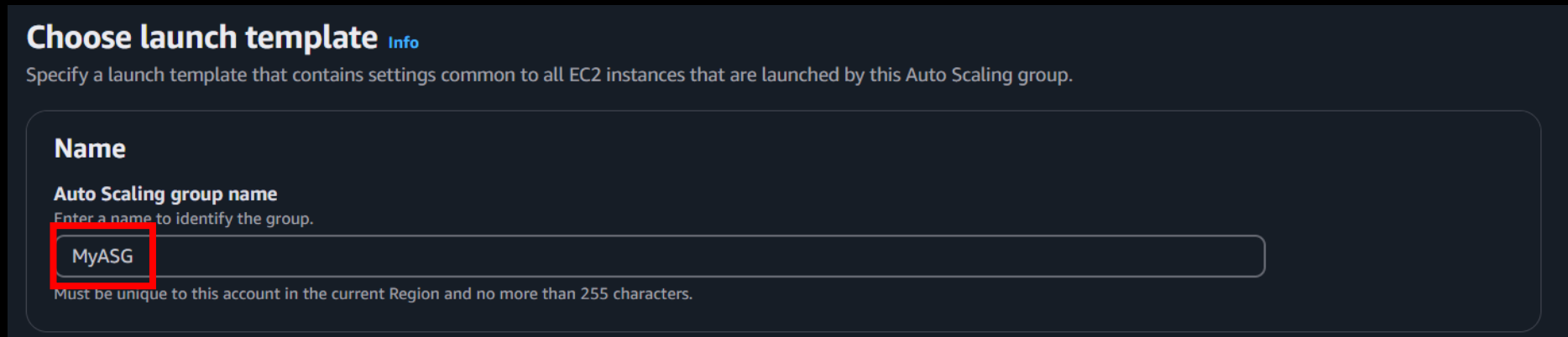
Auto Scaling groups are collections of Amazon EC2 instances that enable automatic scaling and fleet management features. These features help you maintain the health and availability of your applications.

Create Auto Scaling group

Get started with EC2 Auto Scaling by creating an Auto Scaling group

Create Auto Scaling group

3. Provide a name for your ASG.



Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name
Enter a name to identify the group.

MyASG

Must be unique to this account in the current Region and no more than 255 characters.

4. In the launch template selection, choose the previously created launch template.

Launch template [Info](#)

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

Select a launch template ▲

🔍 Search launch templates

MyTemp

↺

Cancel

Next

Launch template [Info](#)

i For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template
Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

MyTemp

[Create a launch template](#)

Version

Default (1)

[Create a launch template version](#)

Description -	Launch template MyTemp lt-0ae32f82c91aa1786	Instance type t2.micro
AMI ID ami-018875e7376831abe	Security groups -	Request Spot Instances No
Key pair name key-win	Security group IDs -	

Additional details

Storage (volumes)	Date created	
--------------------------	---------------------	--

5. Scroll down and click on "Next".

Additional details

Storage (volumes) -	Date created Sun Feb 02 2025 19:55:06 GMT+0530 (India Standard Time)
-------------------------------	--

[Cancel](#) [Next](#)

6. In the Network settings, select the Availability Zones (AZs) and click "Next".

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC
Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-04df38aa883c1f075
172.31.0.0/16 Default ▼

↺

[Create a VPC](#) [↗](#)

Availability Zones and subnets
Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets ▼

↺

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets



us-east-2a | subnet-0a76eaabb79e2d87

172.31.0.0/20 Default



us-east-2b | subnet-08100cbacca947e6d

172.31.16.0/20 Default



us-east-2c | subnet-09cd3cd50e4b75155

172.31.32.0/20 Default

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets ▼



us-east-2c | subnet-09cd3cd50e4b75155 ✕
172.31.32.0/20 Default

us-east-2b | subnet-08100cbacca947e6d ✕
172.31.16.0/20 Default

us-east-2a | subnet-0a76eaabbc79e2d87 ✕
172.31.0.0/20 Default

7. Click “Next”

Availability Zone distribution - *new*

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

☒ **Balanced best effort**
If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

☐ **Balanced only**
If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Cancel

Skip to review

Previous

Next

8. In the Load Balancers section, keep the default setting (No Load Balancer) and click "Next".

Integrate with other services – optional [Info](#)

Use a load balancer to distribute network traffic across multiple servers. Enable service-to-service communications with VPC Lattice. Shift resources away from impaired Availability Zones with zonal shift. You can also customize health check replacements and monitoring.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.



No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.



Attach to an existing load balancer

Choose from your existing load balancers.



Attach to a new load balancer

Quickly create a basic load balancer to attach to your Auto Scaling group.

9. In the "Configure Group Size and Scaling" section, set the **desired capacity** as 2, **Min desired capacity** as 2 & **Max desired capacity** as 10.

Desired capacity
Specify your group size.

2

Scaling [Info](#)
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
2
Equal or less than desired capacity

Max desired capacity
10
Equal or greater than desired capacity

10. Keep everything else same, scroll down and click "Next".

Additional settings

Instance scale-in protection
If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

☐ Enable instance scale-in protection

Monitoring | [Info](#)

☐ Enable group metrics collection within CloudWatch

Default instance warmup | [Info](#)
The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ Enable default instance warmup

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

11. Click "Next" on the 5th and 6th steps as they are optional.

Add notifications - optional [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

[Add notification](#)

[Cancel](#) [Skip to review](#) [Previous](#) [Next](#)

Add tags - optional [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

i You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group. [×](#)

Tags (0)

[Add tag](#)

50 remaining

[Cancel](#) [Previous](#) [Next](#)

12. Review the configurations in the 7th step.

Review Info

Step 1: Choose launch template Edit

Group details
Auto Scaling group name
MyASG

Launch template

Launch template	Version	Description
MyTemp  lt-0ae32f82c91aa1786	Default	

Step 2: Choose instance launch options Edit

Network
VPC
[vpc-04df38aa883c1f075](#) 

Availability Zones and subnets

13. Scroll down and click on "Create Auto Scaling Group".

The screenshot displays the AWS Management Console interface for creating an Auto Scaling group. The breadcrumb navigation at the top indicates the path: **EC2** > **Auto Scaling groups** > **Create Auto Scaling group**. The main configuration area is divided into several sections:

- Capacity Reservation preference**: This section contains three sub-sections: **Preference** (set to Default), **Capacity Reservation IDs** (empty), and **Resource Groups** (empty).
- Step 5: Add notifications**: This section shows **Notifications** as "No notifications". An **Edit** button is located to the right.
- Step 6: Add tags**: This section shows **Tags (0)** with a table header:

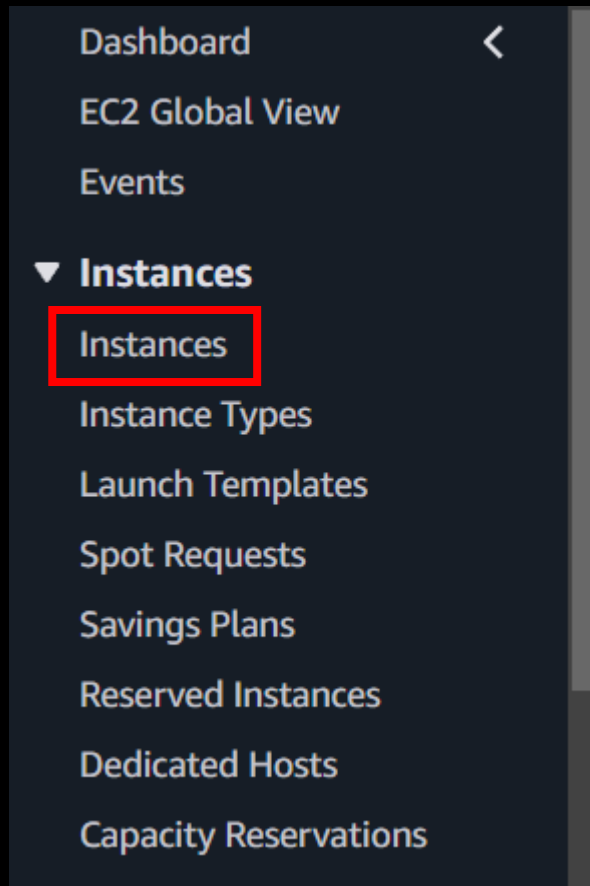
Key	Value	Tag new instances
No tags		

. An **Edit** button is located to the right.

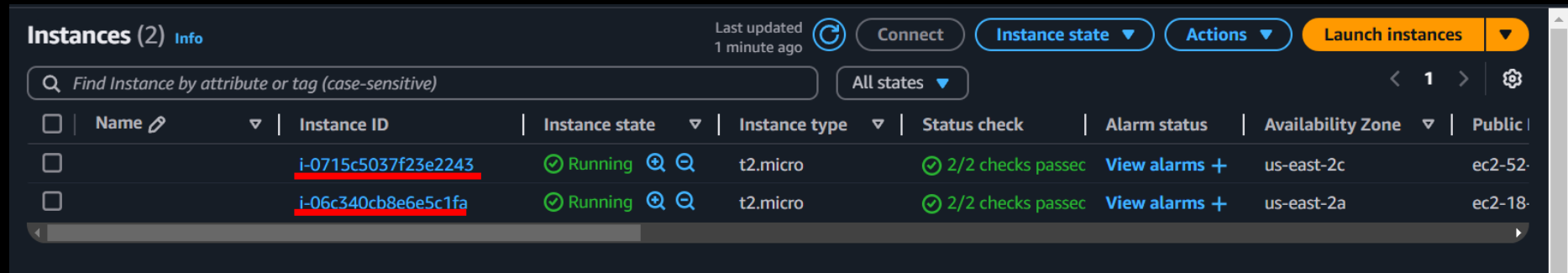
At the bottom of the page, there are three buttons: **Preview code** (with a code icon), **Cancel**, and **Previous**. The **Create Auto Scaling group** button, which is orange, is highlighted with a red rectangle.

IV. VERIFYING AUTO SCALING GROUP ✨

1. Navigate to the EC2 Instance section.



2. Verify that the instances are being created.



The screenshot displays the AWS Management Console 'Instances' page. At the top, it shows 'Instances (2)' with an 'Info' link. A 'Last updated' timestamp of '1 minute ago' is present, along with 'Connect', 'Instance state', 'Actions', and a 'Launch instances' button. A search bar prompts 'Find Instance by attribute or tag (case-sensitive)', and a filter dropdown is set to 'All states'. The instance list table has columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IP. Two instances are listed, both in a 'Running' state with '2/2 checks passed'. The first instance's ID, 'i-0715c5037f23e2243', is highlighted with a red box. The second instance's ID, 'i-06c340cb8e6e5c1fa', is also highlighted with a red box.

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
☐		<u>i-0715c5037f23e2243</u>	Running	t2.micro	2/2 checks passed	View alarms	us-east-2c	ec2-52-
☐		<u>i-06c340cb8e6e5c1fa</u>	Running	t2.micro	2/2 checks passed	View alarms	us-east-2a	ec2-18-

3. Go back to the ASG Dashboard.

The screenshot displays the AWS Auto Scaling Groups dashboard. At the top, there's a header for 'Auto Scaling groups (1/1)' with an 'Info' link. Below this is a search bar and a table of Auto Scaling groups. The table has columns for Name, Launch template/configuration, Instances, Status, Desired capacity, Min, and Max. One group, 'MyASG', is listed with a launch template 'MyTemp' and a version 'Default'. Below the table, there's a section for 'Auto Scaling group: MyASG' with tabs for Details, Integrations - new, Automatic scaling, Instance management, Instance refresh, Activity (highlighted with a red box), and Monitoring. The 'Activity' tab shows the 'MyASG Capacity overview' with an 'Edit' button. The overview includes the ARN, Desired capacity (2), Scaling limits (Min - Max) (2 - 10), Desired capacity type (Units (number of instances)), and Status (-).

Auto Scaling groups (1/1) Info

Search your Auto Scaling groups

Name	Launch template/configuration	Instances	Status	Desired capacity	Min	Max
☒ MyASG	MyTemp Version Default	2	-	2	2	10

Auto Scaling group: MyASG

Details Integrations - new Automatic scaling Instance management Instance refresh **Activity** Monitoring

MyASG Capacity overview Edit

arn:aws:autoscaling:us-east-2:971422709086:autoScalingGroup:96f7f615-4409-499c-b6f3-442673d1e23a:autoScalingGroupName/MyASG

Desired capacity	Scaling limits (Min - Max)	Desired capacity type	Status
2	2 - 10	Units (number of instances)	-

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4. Navigate to "Activity" and compare the Instance IDs.

Status ▾	Description ▾	Cause ▾	Start time ▾	En
✔ Successful	Launching a new EC2 instance: i- <u>0715c5037f23e2243</u>	At 2025-02-02T14:39:47Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2025-02-02T14:39:49Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2025 February 02, 08:09:51 PM +05:30	20 Fe 08 +0
✔ Successful	Launching a new EC2 instance: i- <u>06c340cb8e6e5c1fa</u>	At 2025-02-02T14:39:47Z a user request created an AutoScalingGroup changing the desired capacity from 0 to 2. At 2025-02-02T14:39:49Z an instance was started in response to a difference between desired and actual capacity, increasing the capacity from 0 to 2.	2025 February 02, 08:09:51 PM +05:30	20 Fe 08 +0

Instance ID	Description ▾
<u>i-0715c5037f23e2243</u>	Launching a new EC2 instance: i- <u>0715c5037f23e2243</u>
<u>i-06c340cb8e6e5c1fa</u>	Launching a new EC2 instance: i- <u>06c340cb8e6e5c1fa</u>

V. CLEANING UP RESOURCES

1. Terminate the instances.

Terminate (delete) instances? ×

 On an EBS-backed instance, the default action is for the root EBS volume to be deleted when the instance is terminated. Storage on any local drives will be lost.

Are you sure you want to terminate these instances?

Instance ID	Termination protection
 i-0715c5037f23e2243	 Disabled
 i-06c340cb8e6e5c1fa	 Disabled

To confirm that you want to delete the instances, choose the terminate button below. Instances with termination protection enabled will not be terminated. Terminating the instance cannot be undone.

Cancel Terminate (delete)

✓ Successfully initiated termination (deletion) of i-0715c5037f23e2243,i-06c340cb8e6e5c1fa

Instances (2) Info Last updated less than a minute ago Connect Instance state ▾ Actions ▾ Launch instances ▾

Find Instance by attribute or tag (case-sensitive) All states ▾ < 1 > ⚙️

☐	Name ↗	Instance ID	Instance state ▾	Instance type ▾	Status check	Alarm status	Availability Zone ▾	Public IP
☐		i-0715c5037f23e2243	Shutting-d... 🔍 🔄	t2.micro	-	View alarms +	us-east-2c	ec2-52-
☐		i-06c340cb8e6e5c1fa	Shutting-d... 🔍 🔄	t2.micro	-	View alarms +	us-east-2a	ec2-18-

2. Delete the Auto Scaling Group (ASG).

Auto Scaling groups (1/1) Info ⌂ Launch configurations Launch templates [↗](#) Actions ▲ Create Auto Scaling group

Search your Auto Scaling groups < 1 > ⚙️

☒	Name	Launch template/configuration ↗	Instances ▾	Status	Desired capacity ▾	Min ▾	Max ▾
☒	MyASG	MyTemp Version Default	2	-	2	2	2

Actions Edit **Delete**

Delete Auto Scaling group ✕

i Auto Scaling group contains running instances
Deleting these Auto Scaling groups will terminate all instances in each group.
This action cannot be undone.

Are you sure you want to delete this Auto Scaling group?

- MyASG

Deleting the Auto Scaling group will terminate all instances in the group. This action cannot be undone.

To confirm deletion, type *delete* in the field.

Cancel Delete

Status
⋮ Deleting

3. Delete the Launch Template.

The screenshot displays the AWS Management Console interface for 'Launch Templates'. At the top, there's a header 'Launch Templates (1/1) Info' with a search bar and a 'Create launch template' button. Below this is a table listing launch templates. The first row is selected, showing 'lt-0ae32f82c91aa1786' with the name 'MyTemp' and default version '1'. An 'Actions' dropdown menu is open for this row, with 'Delete template' highlighted. Below the table, the details for 'MyTemp (lt-0ae32f82c91aa1786)' are shown, including a 'Launch template details' section with fields for ID, name, version, and owner. The 'Delete template' button is also visible in the details section.

Launch Template ID	Launch Template Name	Default Version	Created By
lt-0ae32f82c91aa1786	MyTemp	1	arn:aws:iam::971422709086:root

MyTemp (lt-0ae32f82c91aa1786)

Launch template details

Launch template ID	Launch template name	Default version	Owner
lt-0ae32f82c91aa1786	MyTemp	1	arn:aws:iam::971422709086:root

Details | Versions | Template tags

Delete Launch Template



You can't undo this action. Any Auto Scaling groups or Spot Fleet requests currently using this launch template might be affected.

Are you sure you want to delete MyTemp (lt-0ae32f82c91aa1786) and all its versions?

To confirm deletion, type *Delete* in the field

► CLI commands

Cancel

Delete

✓ Delete Launch Template Request Succeeded



Launch Templates [Info](#)



Actions ▼

Create launch template

Q Search

< 1 > ⚙

Launch Template ID ▼	Launch Template Name ▼	Default Version ▼	Latest Version ▼	Create Time ▼	Created By
----------------------	------------------------	-------------------	------------------	---------------	------------

You do not have any Launch Templates in this region

CONCLUSION 🎯

Congratulations 🎉 You have successfully created an Auto Scaling Group (ASG) in AWS, ensuring that your infrastructure can dynamically scale based on demand. This enhances reliability, optimizes costs, and improves overall efficiency. Keep exploring AWS to refine your auto-scaling strategies 🚀

*Efficiency is doing things right; effectiveness is doing
the right things 💡*

– Peter Drucker